



AI-Driven Depression Severity Classification

A Machine-Learning Framework for Eyes-Closed EEG Analysis

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Research Overview

The Challenge: Adolescent Depression

- Global Urgency: Mental health in adolescence is a critical priority.
- Diagnostic Limitations: Address the need for objective innovation.
- Research Motivation: Investigate whether EEG-derived features can support depression-severity classification.



The Signal: Why EEG?

- **Utility**

EEG is a non-invasive physiological measurement that may provide useful research features.

- **Resting-State Dynamics**

Captures intrinsic neural oscillations.

- **Protocol Standardization**

The eyes-closed protocol provides a standardized resting condition and may reduce task-related variability.

Project Overview

Research Goal: Decision-Support Framework

Primary Objective

Develop a research framework for three-class depression-severity classification.

Clinical Philosophy

Designed as a future-facing decision-support application.

Implementation Caveat

Deployment is strictly subject to rigorous external clinical validation.



Data Overview

Dataset derived from the Rachamanee & Wongupparaj (2021) study

Total Participants

85

Feature Density

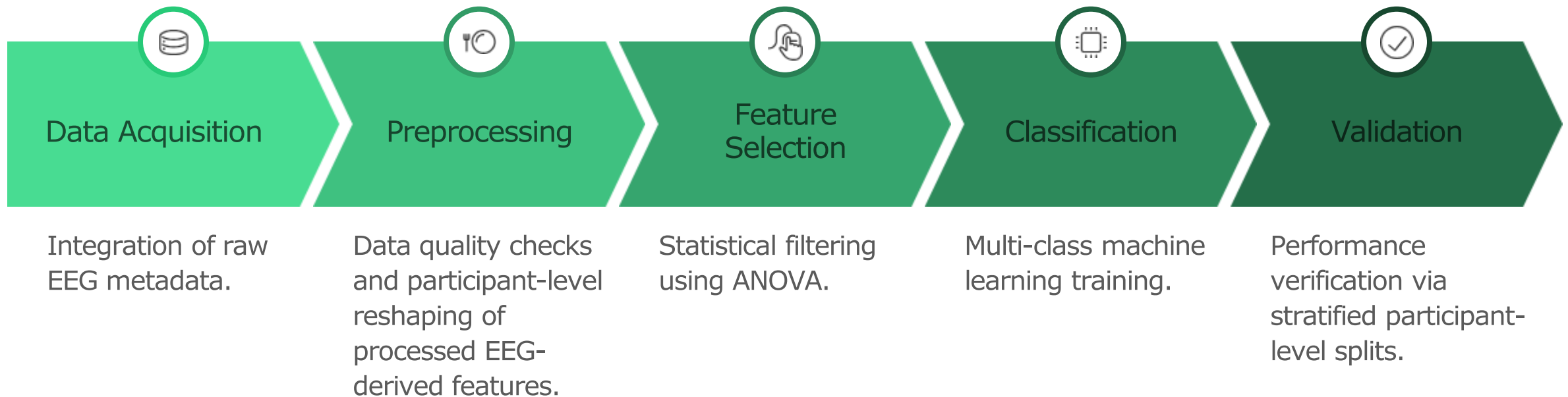
600

features per participant

Includes processed EEG-derived
spectral features.

Reproducible Pipeline Workflow

A structured approach to EEG data processing and analysis



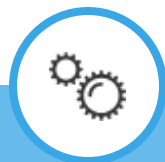
This workflow ensures systematic and reliable EEG data processing from acquisition to final validation.

Design Integrity & Data Leakage Prevention



Isolation Principle

Strict pipeline isolation.



Training Integrity

Preprocessing and feature selection calculated within the training fold.



Test Set Security

The hold-out set remains unseen until final assessment.

Exploratory Data Analysis (EDA)



Data Cleaning

Systematic inspection of missing values and feature-level outliers.



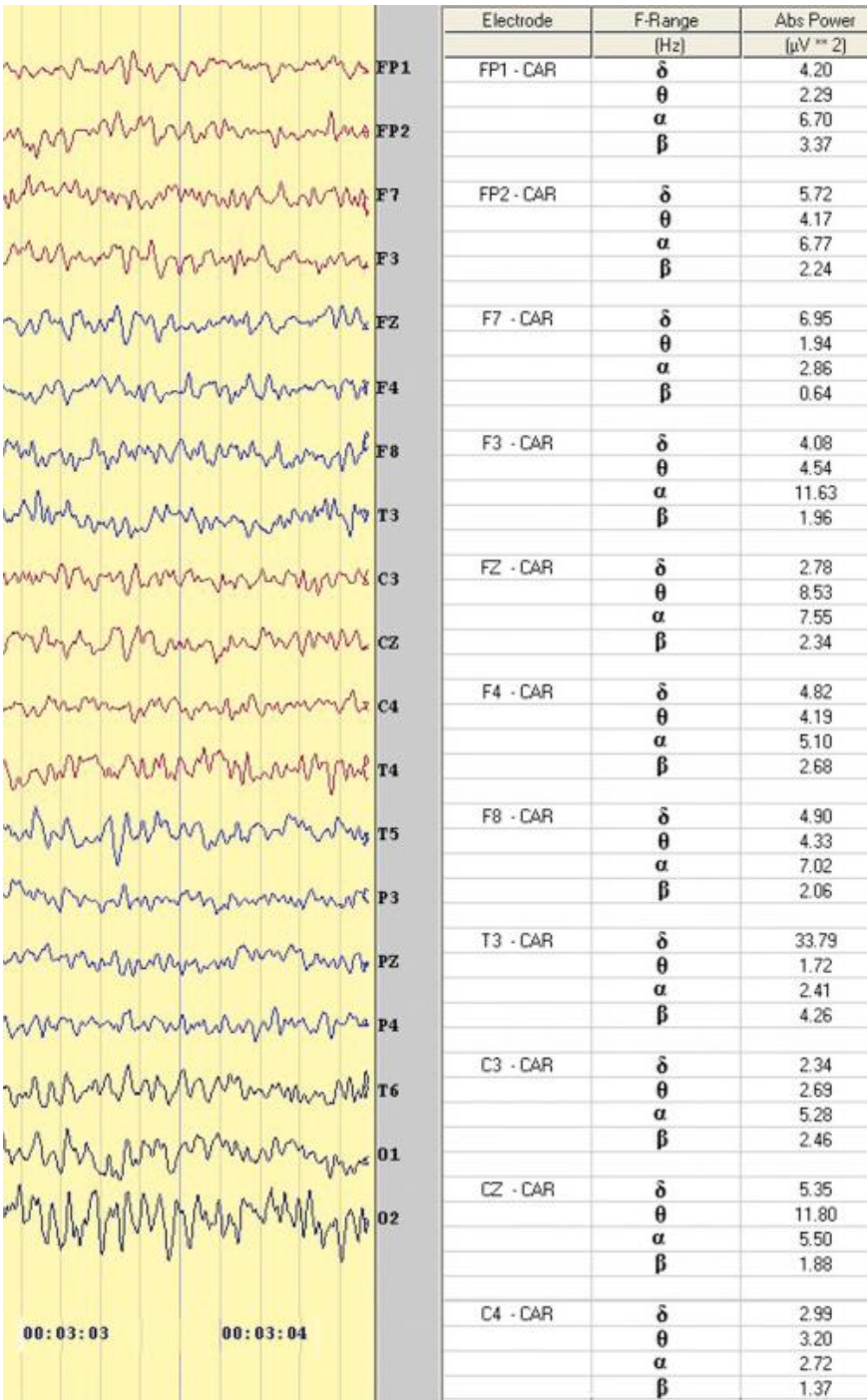
Structural Transformation

Reshaping high-dimensional feature vectors.



Statistical Mapping

Correlation matrices to detect multi-collinearity.



Approach Details

Modeling Methodology

- Feature Matrix (X): Constructed from processed EEG spectral metrics.
- Target (y): 3-class severity scheme.
- Stratification: Participant-level split with 68 training and 17 held-out participants.

Exclusion Criteria & Leakage Control

Data processing and feature integrity measures for predictive modeling



- **Input Feature Constraints**

Focus on processed EEG features.

- **Data Sanitation & Privacy**

Identifiers and demographic variables were excluded from predictive features; Group was retained only as target y .

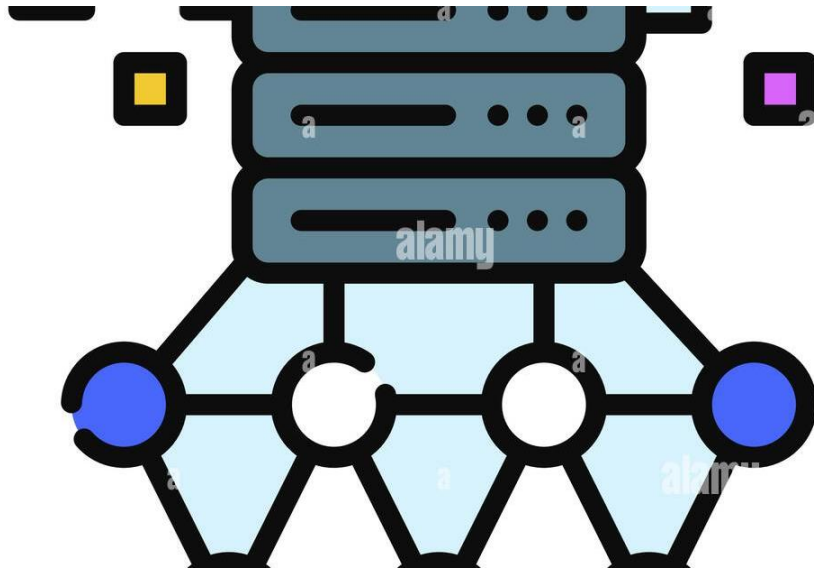
- **Prevention of Target Leakage**

Exclusion of BDI-II and PHQ-9 scores from independent variables.

Evaluation Design

Training and Evaluation Framework

Training Set (68)



Used for 5-fold cross-validation and hyperparameter optimization.

Held-Out Test Set (17)

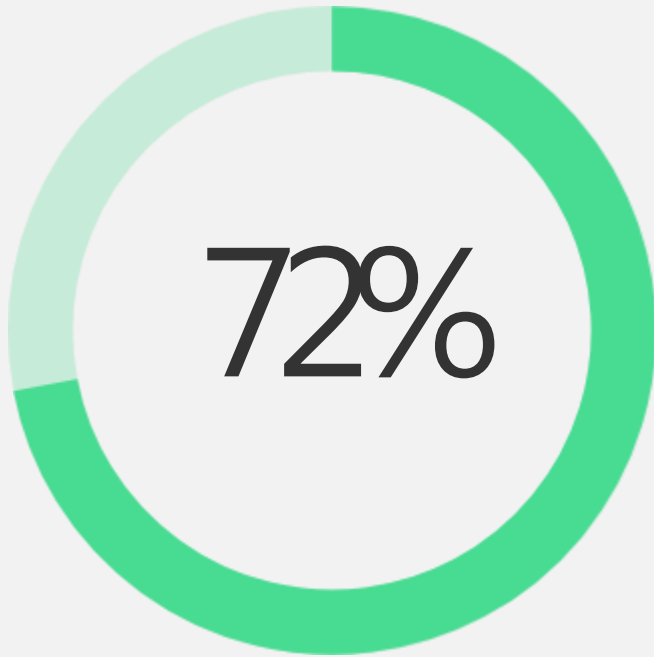


Used as the primary held-out evaluation on unseen participants from the same dataset.

Metrics for Success

Performance evaluation across class categories

Balanced Accuracy



Macro-F1 Score

63.8%

These metrics provide a comprehensive view of model performance, ensuring that no single class dominates the evaluation and that the system remains robust across all categories.

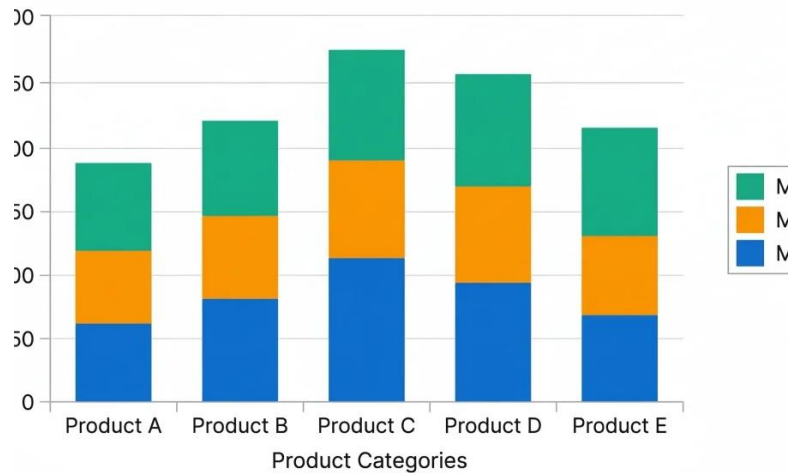
These metrics ensure a more balanced assessment across classes.

Baseline Models Comparison

Evaluation of performance and characteristics

Extra Trees

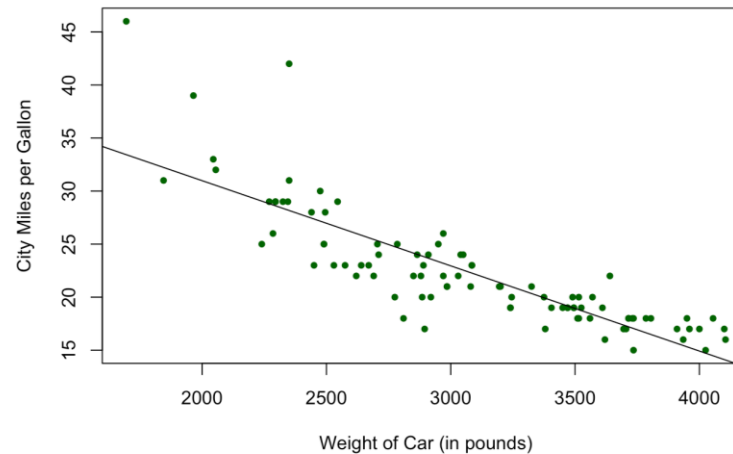
Product Performance across Key Metrics



Strongest baseline performance.

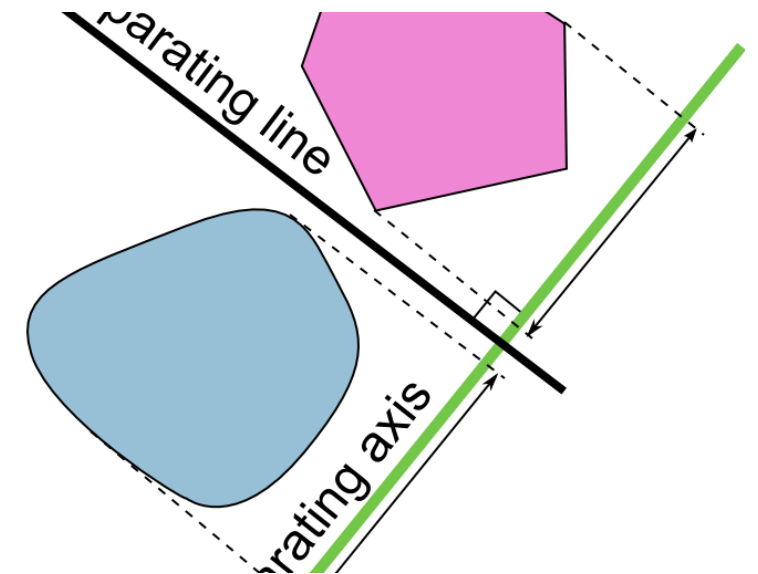
Logistic Regression

Scatterplot of Weight of Car vs City MPG



Linear baseline.

SVM



Non-linear comparison for high-dimensional spaces.

Tuning the Extra Trees Model

Model optimization parameters and feature selection settings



Ensemble Size

Selected 300 trees during hyperparameter tuning.



Feature Strategy

Top 100 features using ANOVA F-score.



Parameter Optimization

Fine-tuning min_samples_leaf.

Rigorous Selection Protocol



Cross-Validation

5-Fold Stratified approach.



Statistical Integrity

Preserving target distribution.



Stability Assessment

Used to quantify performance variability and reduce selection bias.

Held-Out Test Classification Metrics

Performance analysis of participant classification

Accuracy

64.7

Accuracy

Macro-F1 Score

63.8

Macro-F1 Score

Balanced Accuracy

64.4

Balanced Accuracy

Patient Classification

11 out of 17 participants correctly classified

Performance Overview

Class-Wise Performance Analysis

Performance fluctuations were detected across categories.

Interpretation of results should be cautious due to the limited sample size.

Understanding Uncertainty & Sensitivity

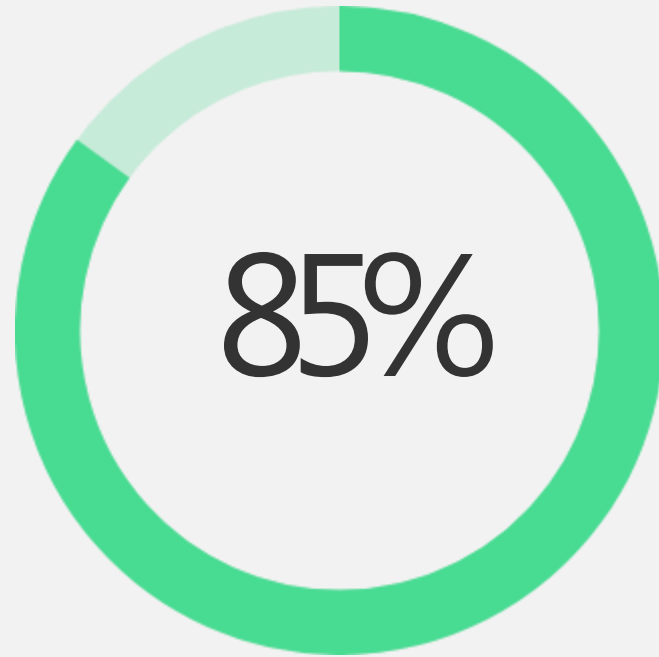
Stability and Variance Analysis in Neuro-imaging Research

Repeated Cross-Validation Stability

0.521

Balanced Accuracy

Variance Sensitivity



Interpretation

High sensitivity to individual subject data due to small-sample neuro-imaging research.

Interpreting Variance: High sensitivity to individual subject data due to small-sample neuro-imaging research.

Transparency in Revision

Evolution of our pipeline and reproducibility standards

Manual/Legacy

- Fragmented pipeline processes
- Heavy reliance on manual intervention
- High susceptibility to human error in calculations

Integrated/Modular

- Transitioned to an integrated, modular pipeline
- Automated preprocessing workflows
- Reduced manual intervention in calculations for improved reproducibility

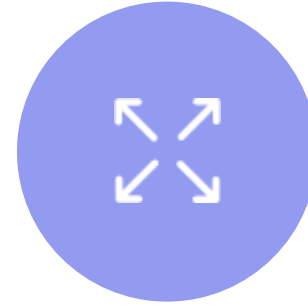
By shifting to a modular and automated approach, we have significantly increased transparency and reliability across our revision processes.

Limitations and Realism



Sample Size Constraints

The dataset contains 85 participants, with 68 used for training, limiting statistical power and generalizability.



Generalization Challenges

Broader validation is required.

Summary

Conclusion: The Path Forward

- Proof of Concept: EEG-derived features show potential for modeling depression severity.
- Vision: Foundational architecture for future clinical research integration.



EEG-derived features offer a promising foundation for future clinical depression severity modeling.

Future Research Directions

Strategic roadmap for advancing EEG research and model reliability.

Research Development Pillars



Scale

Expand cohort size and diversity to enhance generalizability.



Validation

Conduct multi-site external testing to ensure robust performance.



Explainability

Use interpretable EEG feature analysis to improve trust and transparency.

References



- achamanee, S., & Wongupparaj, P. (2021). Resting-state EEG datasets of adolescents with mild, minimal, and moderate depression. BMC Research Notes, 14, 256. <https://doi.org/10.1186/s13104-021-05673-x>
- Acharya et al. (2018)
- Cai et al. (2020)
- Müller & Hegerl (2021)
- Smith et al. (2022)